🔐 Cyber Security Roadmap (Week-wise Structured Plan)

🧠 PHASE 1: Foundations (Networking + OS Basics)

📅 Weeks 1–6

📘 Week 1: Introduction to Cyber Security

- What is Cyber Security?
- Domains:
  - Ethical Hacking
  - Network Security
  - SOC Analyst

📘 Week 2: Computer Basics

- How computers work
- Hardware basics
- OS overview

📘 Week 3: Operating Systems

- Windows basics
- Linux introduction

👉 Install:

- Linux (Ubuntu recommended via VM)

📘 Week 4: Linux Commands (VERY IMPORTANT)

- ls, cd, mkdir, rm
- chmod (permissions)

👉 Practice daily

📓 Week 5: Networking Basics

- What is a network?
- LAN, WAN

📓 Week 6: IP Addressing

- IPv4
- Public vs Private IP

👉 Practice:

- Check your IP
- Basic configs

🔒 PHASE 2: Networking & Security Basics

📅 Weeks 7–14

📓 Week 7: OSI Model

- 7 layers (important)

📓 Week 8: Protocols

- HTTP / HTTPS
- DNS
- TCP / UDP

📓 Week 9: Networking Devices

- Router
- Switch
- Firewall

📓 Week 10: Network Tools

- ping
- traceroute
- netstat

📓 Week 11: Intro to Security

- CIA Triad:
  - Confidentiality
  - Integrity
  - Availability

📓 Week 12: Common Attacks

- Phishing
- Malware
- DDoS

📓 Week 13: Security Tools

- Wireshark basics

📓 Week 14: Practice Platform Start

👉 Start:

- TryHackMe (recommended)

PHASE 3: Ethical Hacking (Core Skills)

Weeks 15–26

Week 15–16: Reconnaissance

- Information gathering
- Tools:
    - Nmap

Week 17–18: Scanning

- Open ports
- Vulnerability scanning

Week 19–20: Exploitation Basics

- What is exploit?
- Basic vulnerabilities

Week 21–22: Web Security Basics

- HTTP requests
- Web structure

Week 23–24: Web Attacks

- SQL Injection
- XSS

📓 Week 25–26: Practice Labs

👉 Platforms:

- TryHackMe
- Hack The Box

🧠 PHASE 4: Advanced Security Skills

📅 Weeks 27–36

📓 Week 27–28: Cryptography

- Encryption / Decryption
- Hashing

📓 Week 29–30: Authentication Systems

- Password security
- Multi-factor authentication

📓 Week 31–32: Network Security

- Firewalls
- IDS/IPS

📓 Week 33–34: System Security

- Hardening systems
- Updates

Week 35–36: Cloud Security Basics

- AWS security
- Permissions

PHASE 5: Tools + Real-World Skills

Weeks 37–44

Week 37–38: Kali Linux

- Security tools
- Environment setup

Week 39–40: Metasploit

- Exploitation framework

Week 41–42: Burp Suite

- Web testing tool

Week 43–44: Bug Bounty Basics

- Finding vulnerabilities
- Reporting bugs

PHASE 6: Projects & Practice

Weeks 45–50

📓 Week 45–46: Beginner Projects

- Network scanning
- Basic security audit

📓 Week 47–48: Intermediate Projects

- Web vulnerability testing
- Secure system setup

📓 Week 49–50: Advanced Project

👉 Perform:

- Full penetration testing simulation

🧠 PHASE 7: Job Preparation

📅 Weeks 51–52

📓 Week 51: Resume + Portfolio

- Show:
  - Labs completed
  - Skills

📓 Week 52: Apply + Certifications

- Apply for:
  - SOC Analyst
  - Junior Security roles

👉 Certifications:

- CEH (optional)

- CompTIA Security+

Weekly Routine

- 5 days learning
- 2 days hands-on labs

Daily:

- 2–4 hours

Final Outcome (After 1 Year)

By Week 52:

✅ Strong networking knowledge
✅ Linux mastery
✅ Ethical hacking skills
✅ Hands-on lab experience
✅ Security tools knowledge

👉 Job Roles:

- Ethical Hacker
- SOC Analyst
- Security Analyst